Statements, Implications, and Negations

MTH 317H — Week 3

Name: _____

1 Warm-up: proving equality of sets

Let

$$A = \{x^2 : x \in \mathbb{R}\}, \qquad B = \{y \in \mathbb{R} : y \geq 0\}, \qquad C = \{x^3 : x \in \mathbb{R}\}.$$

Preparation

To prove that two sets are equal, prove both containments:

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Proof

Why $A = B$.
First, let $a \in A$. Then $a = x^2$ for some $x \in \mathbb{R}$. Since every real square is nonnegative, $a \geq 0$, so $a \in B$. Therefore $A \subseteq B$.
Conversely, let $b \in B$. Then $b \geq 0$, so $\sqrt{b} \in \mathbb{R}$ and

$$(\sqrt{b})^2 = b.$$

Thus $b \in A$, and hence $B \subseteq A$. We conclude that $A = B$.

Comparing B and C.
The sets B and C are not equal. For example,

$$-1 = (-1)^3 \in C, \qquad \text{but} \qquad -1 \notin B.$$

In fact, every real number has a real cube root, so $C = \mathbb{R}$.

2 Statements and logical connectives

Definition 2.1. A *statement* is a sentence that is either true or false. The sentence does not have to be mathematical.

Examples of statements include:

- "It snowed yesterday."
- "$\mathbb{R}$ is a vector space."
- "The equation $3x - 7 = 0$ has no real solutions."

The third sentence is false, but it is still a statement because it has a well-defined truth value.

The following are not statements:

- "What are the roots of $x^2 - 2x + 1$?"
- "Assume $y \in \mathbb{Z}$."

The first is a question, and the second is an instruction; neither is true or false on its own.

2.1 Conjunction and disjunction

If P and Q are statements, then we can form new statements using *and* and *or*:

$$P \wedge Q \quad (\text{"}P \text{ and } Q\text{"}), \qquad P \vee Q \quad (\text{"}P \text{ or } Q\text{"}).$$

For example, let

$$P = \text{"the set } X \text{ is finite"}, \qquad Q = \text{"the set } X \text{ is closed under addition"}.$$

Then $P \wedge Q$ asserts both properties, while $P \vee Q$ asserts that at least one of the two properties holds. Here, as usual in mathematics, "or" is inclusive: $P \vee Q$ is also true when both statements are true.

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

3 Quantified statements

The symbols $\forall$ and $\exists$ allow us to state how many objects have a given property:

$$\forall \text{ means "for every"}, \qquad \exists \text{ means "there exists"}.$$

Example 3.1 (A universal statement)**.** The sentence

$$\forall x \in \mathbb{R}, \qquad f'(x) = 0$$

for the function $f(x) = 17x$ is false. Indeed, $f'(x) = 17$ for every x. To disprove a universal statement, it is enough to find one counterexample.

Example 3.2 (Another universal statement)**.** Consider

$$\forall x \in \mathbb{R}, \qquad f'(x) > 0$$

for $f(x) = 2x^2 - 7x + 3$. Since $f'(x) = 4x - 7$, taking $x = 0$ gives $f'(0) = -7 < 0$. The statement is therefore false.

Example 3.3 (An existential statement)**.** The sentence "there exists a positive real number whose square is 5" can be written as

$$\exists x \in \mathbb{R} \quad \text{such that} \quad x > 0 \text{ and } x^2 = 5.$$

It is true: choose $x = \sqrt{5}$.

Preparation

To disprove a statement beginning with $\forall$, provide a counterexample. To prove a statement beginning with $\exists$, provide a witness and verify that it has the required properties.

4 Negations

Definition 4.1. If P is a statement, its *negation* is written $\neg P$ and means "not P." A statement and its negation have opposite truth values.

Preparation

The fundamental negation rules are

$$\begin{aligned} \neg(P \wedge Q) &\equiv (\neg P) \vee (\neg Q), \\ \neg(P \vee Q) &\equiv (\neg P) \wedge (\neg Q), \\ \neg(\forall x \in S,\ P(x)) &\equiv \exists x \in S \text{ such that } \neg P(x), \\ \neg(\exists x \in S \text{ such that } P(x)) &\equiv \forall x \in S,\ \neg P(x). \end{aligned}$$

Example 4.2 (Negating a conjunction)**.** Let

$$P: \qquad \mathbb{N} \subseteq \mathbb{Z} \ \wedge\ \mathbb{N} \subseteq \mathbb{R}.$$

Then

$$\neg P: \qquad \mathbb{N} \not\subseteq \mathbb{Z} \ \vee \ \mathbb{N} \not\subseteq \mathbb{R}.$$

The original statement is true, so its negation is false.

Example 4.3 (Negating alternating quantifiers)**.** Let

$$P: \qquad \exists y \in \mathbb{Z} \text{ such that } \forall x \in \mathbb{Z}, \ x + y = 1.$$

Its negation is

$$\neg P: \qquad \forall y \in \mathbb{Z}, \ \exists x \in \mathbb{Z} \text{ such that } x + y \neq 1.$$

The original statement is false. In fact, for any proposed $y \in \mathbb{Z}$, choose $x = 2 - y$. Then $x \in \mathbb{Z}$ and $x + y = 2 \neq 1$, which proves the negation.

Example 4.4 (Negating a statement with "or")**.** Let

$$P: \qquad \forall x \in \mathbb{R}, \quad x^2 > 1 \ \vee \ -1 \leq x \leq 1.$$

Using De Morgan's law and reversing the inequalities, the negation is

$$\neg P: \qquad \exists x \in \mathbb{R} \text{ such that } x^2 \leq 1 \ \wedge \ (x < -1 \ \vee \ x > 1).$$

The original statement is true: every real number either lies in $[-1, 1]$ or has absolute value greater than 1, in which case its square is greater than 1.

5 Implications and biconditionals

Definition 5.1. An *implication* has the form

$$\text{if } P \text{ is true, then } Q \text{ is true,} \qquad \text{written} \qquad P \Rightarrow Q.$$

We say that P *implies* Q. The implication is false exactly when P is true and Q is false.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Example 5.2 (Divisibility)**.** For $x \in \mathbb{Z}$, consider

$$8 \mid x \quad \Rightarrow \quad 4 \mid x.$$

Proof

If $8 \mid x$, then $x = 8n$ for some $n \in \mathbb{Z}$. Hence $x = 4(2n)$, so $4 \mid x$. The reverse implication is false: $4 \mid 4$, but $8 \nmid 4$.

Example 5.3 (Polynomial spaces)**.** Let $\mathcal{P}_n$ denote the vector space of real polynomials of degree at most n. Then

$$f \in \mathcal{P}_2 \quad \Rightarrow \quad f \in \mathcal{P}_3,$$

Proof

because a polynomial of degree at most 2 also has degree at most 3.

Definition 5.4. The statement "P if and only if Q" is the *biconditional*

$$P \iff Q.$$

It abbreviates the conjunction of the two implications:

$$(P \Rightarrow Q) \wedge (Q \Rightarrow P).$$

Thus a proof of an "if and only if" statement must prove both directions.

Summary

Goal	**Typical method**
Prove $A = B$	Prove $A \subseteq B$ and $B \subseteq A$.
Disprove $\forall x \in S,\ P(x)$	Find one $x \in S$ for which $P(x)$ is false.
Prove $\exists x \in S,\ P(x)$	Give a witness x and verify $P(x)$.
Negate a quantified statement	Switch $\forall$ with $\exists$ and negate the property.
Prove $P \iff Q$	Prove both $P \Rightarrow Q$ and $Q \Rightarrow P$.